

Name: _____ Date: _____

Period: _____

Primary Objective**I will be able to...**

- LO 4.3.A** — Identify the direction of planar motion at any value of the parameter by determining whether $x(t)$ and $y(t)$ are increasing or decreasing at that moment. **determine direction from whether each component is increasing or decreasing**
- LO 4.3.A** — Explain why the same curve can be traversed in different directions by different parametrizations. **different parametrizations can produce the same (x, y) pairs in different orders**
- LO 4.3.A** — Compute the average rate of change of $x(t)$ and $y(t)$ over $[t_1, t_2]$ and use their ratio to find the slope. **slope = $\Delta y / \Delta x$ where $\Delta x = x(t_2) - x(t_1)$ and $\Delta y = y(t_2) - y(t_1)$**

Vocabulary & Key Concepts

direction of motion: Determined by whether each component is **increasing** or **decreasing**. If $x(t)$ is increasing $\Rightarrow$ moving **right**; if decreasing $\Rightarrow$ moving **left**. If $y(t)$ is increasing $\Rightarrow$ moving **up**; if decreasing $\Rightarrow$ moving **down**.

ARC of $x(t)$: The average rate of change of $x(t)$ over $[t_1, t_2]$: $\frac{x(t_2) - x(t_1)}{t_2 - t_1}$

ARC of $y(t)$: The average rate of change of $y(t)$ over $[t_1, t_2]$: $\frac{y(t_2) - y(t_1)}{t_2 - t_1}$

slope of parametric curve: The slope of the **secant** line connecting two points on the curve: $\frac{\Delta y}{\Delta x} = \frac{\text{ARC of } y(t)}{\text{ARC of } x(t)}$ provided ARC of $x(t) \neq 0$.

reparametrization: A different pair of component functions that traces the **same** set of points in the plane, possibly in a **different** direction.

Hook

Entry Question: Two drones follow the same figure-eight path — Drone A clockwise, Drone B counterclockwise.

Q1. Can two different parametric functions trace the same path? What would have to be different about them?

Yes. The component functions would be different, producing the same set of (x, y) points but at different values of t , possibly traversed in a different order (direction).

Q2. How do you determine which direction a particle is moving horizontally at a specific value of t ?

Compare $x(t)$ at values slightly before and after the given t . If x is increasing, the particle moves right; if decreasing, left.

LO 4.3.A — Direction of Planar Motion (EK 4.3.A.1–A.2)

Key rule: Analyze $x(t)$ and $y(t)$ *independently*.

If $x(t)$ is increasing	particle moves right
If $x(t)$ is decreasing	particle moves left
If $y(t)$ is increasing	particle moves up
If $y(t)$ is decreasing	particle moves down

Example: Let $f(t) = (t^2 - 2t, 3t - t^2)$ for $0 \leq t \leq 4$.

Step 1 — Build a table of values.

t	$x(t) = t^2 - 2t$	$y(t) = 3t - t^2$	(x, y)
0	0	0	(0, 0)
1	-1	2	(-1, 2)
2	0	2	(0, 2)
3	3	0	(3, 0)
4	8	-4	(8, -4)

Step 2 — Direction at $t = 0.5$.

Compare $x(0)$ and $x(1)$: $x(0) = 0$, $x(1) = -1$.

Since $x(1) < x(0)$, the function $x(t)$ is **decreasing** near $t = 0.5$.

$\Rightarrow$ particle is moving **left**

Compare $y(0)$ and $y(1)$: $y(0) = 0$, $y(1) = 2$.

Since $y(1) > y(0)$, the function $y(t)$ is **increasing** near $t = 0.5$.

$\Rightarrow$ particle is moving **up**

Step 3 — Direction at $t = 2.5$.

Compare $x(2)$ and $x(3)$: $x(2) = 0$, $x(3) = 3$; is x increasing or decreasing? **increasing**

$\Rightarrow$ particle is moving **right**

Compare $y(2)$ and $y(3)$: $y(2) = 2$, $y(3) = 0$; is y increasing or decreasing? **decreasing**

$\Rightarrow$ particle is moving **down**

Teacher Note

Emphasize that direction at $t = 0.5$ and $t = 2.5$ are opposite in both components, even though the particle may be at the same (x, y) point at some instant. This is EK 4.3.A.2: direction can differ for different values of t even at the same location.

LO 4.3.A — Same Curve, Different Parametrizations (EK 4.3.A.3)

Key idea: The same set of points in the plane can be traced by different parametric functions — possibly in different directions.

Example: Let $f(t) = (t, t^2)$ and $g(t) = (3 - t, (3 - t)^2)$ for $0 \leq t \leq 3$.

	Robot A: $f(t) = (t, t^2)$		Robot B: $g(t) = (3-t, (3-t)^2)$	
t	(x, y) for f	direction	(x, y) for g	direction
0	(0, 0)	right, up	(3, 9)	left, down
1	(1, 1)	right, up	(2, 4)	left, down
2	(2, 4)	right, up	(1, 1)	left, down
3	(3, 9)	right, up	(0, 0)	left, down

Do f and g trace the same set of points? **Yes**

Are they moving in the same direction? **No — opposite directions**

LO 4.3.A — Slope of the Curve via Average Rates of Change (EK 4.3.A.4)

Formula: Over the interval $[t_1, t_2]$:

$$\text{slope of curve} = \frac{\Delta y}{\Delta x} = \frac{y(t_2) - y(t_1)}{x(t_2) - x(t_1)} \quad (\text{provided } x(t_2) \neq x(t_1))$$

Example (continued): Use $f(t) = (t^2 - 2t, 3t - t^2)$. Find the slope of the curve from $t = 1$ to $t = 3$.

$$\begin{array}{lll} x(1) = -1 & x(3) = 3 & \Delta x = 4 \\ y(1) = 2 & y(3) = 0 & \Delta y = -2 \end{array}$$

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{-2}{4} = -\frac{1}{2}$$

Interpretation: The secant line connecting the points on the curve at $t = 1$ and $t = 3$ has slope $-\frac{1}{2}$.

Teacher Note

Stress that $\Delta y/\Delta x$ uses the *changes in the output coordinates*, not $\Delta y/\Delta t$ or $\Delta x/\Delta t$ individually. The Δt cancels when you form the ratio of the two ARCs. Students commonly confuse these.